

Project Title:

Analyzing Adverse Drug Events Using OpenFDA Data

Project Idea Description:

For my capstone project, I propose to explore and analyze data from the **OpenFDA Adverse Event Reporting System (FAERS)**. This dataset is publicly provided by the U.S. Food and Drug Administration and contains millions of reports on adverse drug reactions submitted by healthcare professionals, consumers, and pharmaceutical companies.

The core problem this project aims to address is the identification of patterns and trends in reported negative drug reactions. Adverse drug events can have serious public health implications, and analyzing this data can help surface important insights, such as

- Which drugs are most frequently associated with adverse reactions
- What kinds of reactions are most common
- Whether certain demographics (age, gender) are more affected by specific drugs

This dataset provides a rich and complex foundation for analysis and is particularly suitable for a data engineering project due to its size and semi-structured JSON format.

In addition, this topic connects well with my background in software development and my interest in healthcare data, making it both a technically and personally meaningful choice for the capstone.